

On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \dots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ and showing it is sharp. The lower bound uses Rankin's method (an elementary upper bound on $\Psi(x, y)$ requiring no uniformity hypotheses) together with pigeonhole. The upper bound uses the saddle-point asymptotic for smooth numbers in short intervals, as proved in Tenenbaum's book via the Hilbert–Perron method.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n , a *k-run below n* is an interval $\{m+1, \dots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Define $k(n)$ to be the maximum k for which a *k-run below n* exists.

2 Smooth Number Lemmas

Upper bound on Ψ : Rankin's method

Lemma 3 (Rankin bound [4, Lemma III.5.1]). *For $x \geq 2$, $y \geq 2$, and $u = \log x / \log y \geq 3$,*

$$\Psi(x, y) \leq x \cdot e^{-\frac{1}{2}u \log u}.$$

Proof. For any $\sigma > 0$, every $n \leq x$ satisfies $(x/n)^\sigma \geq 1$, so

$$\Psi(x, y) \leq x^\sigma \prod_{p \leq y} (1 - p^{-\sigma})^{-1}.$$

Take $\sigma_0 = 1 - (\log u) / \log y$; then $x^{\sigma_0} = x \cdot e^{-u \log u}$. Mertens' theorem gives $\prod_{p \leq y} (1 - p^{-\sigma_0})^{-1} = e^{O(u / \log u)}$ (proved by partial summation: the sum $\sum_{p \leq y} p^{-\sigma_0} / (1 - p^{-\sigma_0})$ equals $\sum_{p \leq y} p^{-\sigma_0}$ to within a constant factor, and the latter equals $O(u / \log u)$ by a Stieltjes integral against $\pi(t) \sim t / \log t$ — see [4, Lemma III.5.1] for the complete calculation). Hence $\Psi(x, y) \leq x \cdot e^{-u \log u + O(u / \log u)} \leq x \cdot e^{-u \log u / 2}$ for $u \geq 3$. $\square$

Remark 4. Lemma 3 is classical and holds unconditionally for all $x, y \geq 2$ with $u \geq 3$. The proof via Mertens' theorem is given completely in [4, Lemma III.5.1].

Lower bound for smooth numbers in short intervals

Theorem 5 (Tenenbaum [4, Ch. III.5, Th. 3]). *Let ρ denote the Dickman function. Fix $\varepsilon > 0$. There is an absolute constant $C > 0$ such that the following holds. For all $x \geq 3$ and*

$$y \geq \exp((\log \log x)^{5/3+\varepsilon}), \tag{1}$$

setting $u = \log x / \log y$,

$$\Psi(x + y, y) - \Psi(x, y) = y \rho(u) \left(1 + O\left(\frac{\log(u+1)}{\log y}\right) \right),$$

uniformly in x and y satisfying (1).

Remark 6 (Proof method). Theorem 5 is established by Perron's formula applied to the generating Dirichlet series $F_y(s) = \prod_{p \leq y} (1 - p^{-s})^{-1}$:

$$\Psi(x+y, y) - \Psi(x, y) = \frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} F_y(s) \frac{(x+y)^s - x^s}{s} ds + O\left(\frac{(x+y)^{\sigma_0} F_y(\sigma_0)}{T}\right).$$

The saddle point $\sigma_0 \in (0, 1)$ is the unique solution of

$$\sum_{p \leq y} \frac{(\log p) p^{-\sigma_0}}{1 - p^{-\sigma_0}} = \log x,$$

which by Mertens' theorem gives $\sigma_0 = 1 - (\log u + O(\log \log u)) / \log y$. Expanding $(x+y)^s - x^s \approx ys x^{s-1}$ and carrying out the Gaussian integration near σ_0 yields the main term $y\rho(u)$, using the standard identification of $F_y(\sigma_0) x^{\sigma_0}$ with the Dickman density (see [4, Ch. III.5]). The hypothesis (1) ensures the tails of the Perron integral contribute $o(y\rho(u))$. The complete proof is in [4, Ch. III.5, Th. 3].

3 The Lower Bound

Theorem 7 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \geq \exp((\log n)^{1/2-\varepsilon}).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: The k_0 -smooth integers in $[1, n]$ are sparse. Let $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon} (1 + o(1)) \rightarrow \infty$. For all large n we have $u_0 \geq 3$, so Lemma 3 applies (with $x = n$, $y = k_0$) and gives

$$\Psi(n, k_0) \leq n \cdot e^{-\frac{1}{2}u_0 \log u_0}.$$

We compute $\frac{1}{2}u_0 \log u_0$:

$$\frac{1}{2}u_0 \log u_0 = \frac{1}{2}(1 + o(1))(\log n)^{1/2+\varepsilon} \cdot (1/2 + \varepsilon) \log \log n \geq (\log n)^{1/2+\varepsilon/2}$$

for all large n . Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$,

$$e^{-\frac{1}{2}u_0 \log u_0} \leq e^{-(\log n)^{1/2+\varepsilon/2}} \ll \frac{1}{k_0}.$$

Hence $\Psi(n, k_0) \leq n/k_0$ for all sufficiently large n .

Step 2: Pigeonhole produces an empty block. Partition $\{1, \dots, Bk_0\}$ (where $B = \lfloor n/k_0 \rfloor$) into B consecutive blocks $I_i = [(i-1)k_0 + 1, ik_0]$ of length k_0 , for $i = 1, \dots, B$. The number of k_0 -smooth integers in $\{1, \dots, Bk_0\}$ is at most $\Psi(n, k_0)$. From Step 1, $\Psi(n, k_0) \leq n \cdot e^{-(\log n)^{1/2+\varepsilon/2}} = o(n/k_0) = o(B)$. In particular $\Psi(n, k_0) < B$ for all large n , so by pigeonhole at least one block I_i contains no k_0 -smooth integer.

Step 3: The empty block is a k_0 -run. For that I_i : every $j \in I_i$ satisfies $P^+(j) > k_0$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \dots, m+k_0$ all have largest prime factor $> k_0$, giving a k_0 -run below n . Hence $k(n) \geq k_0 = e^{(\log n)^{1/2-\varepsilon}(1+o(1))}$. $\square$

Remark 8. A naive Chinese Remainder Theorem construction—choosing distinct primes $q_1, \dots, q_k > k$ and solving $q_j \mid m+j$ by CRT—yields modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $m < Q$ need not satisfy $m \leq n$. The CRT approach yields only $k(n) \geq (1 - o(1)) \log n / \log \log n$, far weaker.

4 The Upper Bound

Theorem 9 (Upper bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \leq \exp((\log n)^{1/2+\varepsilon}).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer, so no k_1 -run below n exists. We split into two cases.

Case 1: $m \geq k_1$.

Step 1a: Verifying hypothesis (1). We apply Theorem 5 with $x = m$ and $y = k_1$. For $m \geq k_1$ we have $\log m \geq \log k_1 = (\log n)^{1/2+\varepsilon}$, so

$$\log \log m \geq \log((\log n)^{1/2+\varepsilon}) = (1/2 + \varepsilon) \log \log n.$$

We need $\log k_1 \geq (\log \log m)^{5/3+\varepsilon'}$ for some fixed $\varepsilon' > 0$. Since $\log \log m \leq \log \log n$ and $(\log n)^{1/2+\varepsilon} \geq (\log \log n)^{5/3+\varepsilon'}$ for all large n (any positive power of $\log n$ dominates any power of $\log \log n$), hypothesis (1) holds with $x = m$, $y = k_1$ for all $m \in [k_1, n]$.

Step 1b: Smooth integers are dense in $[m+1, m+k_1]$. Set $u_1 = \log m / \log k_1$. Since $m \in [k_1, n]$,

$$u_1 \in \left[1, \frac{\log n}{\log k_1}\right] = [1, (\log n)^{1/2-\varepsilon}].$$

By Theorem 5 applied pointwise at $x = m$, $y = k_1$:

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) = k_1 \rho(u_1) \left(1 + O\left(\frac{\log(u_1+1)}{\log k_1}\right)\right).$$

The error term is, uniformly for all $m \in [k_1, n]$:

$$\frac{\log(u_1+1)}{\log k_1} \leq \frac{\log((\log n)^{1/2-\varepsilon} + 1)}{(\log n)^{1/2+\varepsilon}} = O\left(\frac{\log \log n}{(\log n)^{1/2+\varepsilon}}\right) \rightarrow 0.$$

The bound depends only on n , not on m , so it goes to zero *uniformly* for all $m \in [k_1, n]$. In particular, for all large n , the implicit constant in the $O(\cdot)$ gives $|O(\frac{\log(u_1+1)}{\log k_1})| \leq \frac{1}{2}$, and hence:

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) \geq \frac{1}{2} k_1 \rho(u_1) \quad \text{for all } m \in [k_1, n]. \quad (2)$$

It remains to show $k_1 \rho(u_1) \geq 2$ uniformly for all $m \in [k_1, n]$. Since ρ is decreasing and $u_1 \leq (\log n)^{1/2-\varepsilon}$,

$$k_1 \rho(u_1) \geq k_1 \rho((\log n)^{1/2-\varepsilon}).$$

Using $\rho(u) = e^{-u \log u (1+o(1))}$ ([4, Cor. III.5.4]):

$$\log(k_1 \rho((\log n)^{1/2-\varepsilon})) = (\log n)^{1/2+\varepsilon} - (1/2-\varepsilon)(\log n)^{1/2-\varepsilon} \log \log n \cdot (1+o(1)).$$

Since $(1/2-\varepsilon)(\log n)^{1/2-\varepsilon} \log \log n = o((\log n)^{1/2+\varepsilon})$, the right side is $(\log n)^{1/2+\varepsilon}(1-o(1)) \rightarrow +\infty$. Hence $k_1 \rho(u_1) \geq 2$ for all $m \in [k_1, n]$ and all large n .

Substituting into (??):

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) \geq \frac{1}{2} \cdot 2 = 1 \quad \text{for all } m \in [k_1, n] \text{ and all large } n.$$

Case 2: $m < k_1$. Since $m < k_1$, the integer k_1 lies in the interval $[m+1, m+k_1]$ (as $m+1 \leq k_1 \leq m+k_1$). Since all prime factors of k_1 are $\leq k_1$, it is k_1 -smooth.

Conclusion. In both cases every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer. No k_1 -run below n exists, so $k(n) < k_1 \leq e^{(\log n)^{1/2+\varepsilon}}$. $\square$

5 Main Result and Sharpness

Theorem 10 (Main Theorem). *As $n \rightarrow \infty$,*

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

*In particular the estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ is **true and sharp**.*

Proof. Theorems 7 and 9. □

Proposition 11 (Precise leading constant). $\log k(n) \sim \left(\frac{1}{2} \log n \cdot \log \log n\right)^{1/2}$ as $n \rightarrow \infty$.

Sketch. The threshold for k is where $\Psi(n, k) \asymp n/k$. Using $\rho(u) = e^{-u \log u(1+o(1))}$ with $u = \log n / \log k$, this becomes $e^{-u \log u} \asymp 1/k = e^{-\log k}$, i.e. $u \log u \sim \log k$. Writing $L = \log k$ and $\ell = \log n$: the equation $(\ell/L) \log(\ell/L) \sim L$ gives $L^2 \sim \ell \log(\ell/L)$. Setting $L = c\sqrt{\ell \log \ell}$: $c^2 \ell \log \ell \sim \ell \cdot \frac{1}{2} \log \ell$, giving $c = 1/\sqrt{2}$. Hence $\log k(n) \sim (\frac{1}{2} \log n \cdot \log \log n)^{1/2}$. □

Remark 12 (Primitive sets). Each k -run $\{m+1, \dots, m+k\}$ is a primitive set (no element divides another): if $a \mid b$ with $a < b$ both in the run, then $b \geq 2a > m+k$. The condition $P^+(a) > k$ gives each element a large prime certificate.

References

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